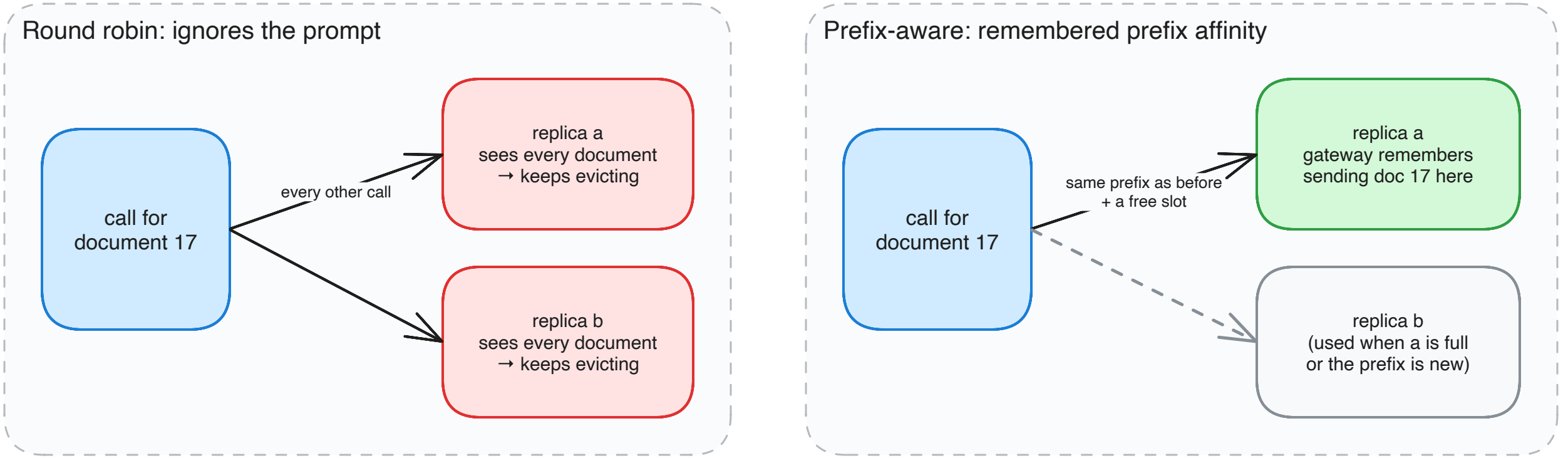
Where should each document go?

40 different 10K-token documents: too many for one replica's KV cache, but about half fit on each

a document call · cache keeps evicting · prefix reused



The gateway does not ask vLLM what is cached: it remembers which prompt prefixes it sent where, and prefers that replica while it has a free slot. Otherwise it places by load.

Measured (goodput · share of prompt served from cache)

Spare capacity, 12 users: round robin 3.70/s · 49% → prefix-aware v2 5.24/s · 91% (+42%)

Saturated, 24 users: prefix-aware v1 4.50/s · 47% → v2 6.12/s · 66% (+36%)